

Best Measure of Center Court

Unit 8: Sports Analytics Lab • Lesson 8-4 • EduWonderLab

UNIT 8 • LESSON 8-4 • TEACHER NOTES

Standard 6.SP.5d

FORMAT Debate / justify	TIME 25–30 min
GROUPING Pairs or teams	TOPIC Appropriate Measures
STANDARD 6.SP.5d	PREP LEVEL Light — print data-set evidence cards and verdict cards

Learning goal *I can decide whether the mean or median better describes a data set and justify it.*

Teacher setup

- Print one card set per pair. Each card has a data set, some with outliers.
- Students decide if the mean or median is the better measure of center.
- They use evidence from shape, outliers, and context.

Materials

Data-set evidence cards, Court argument organizer, Verdict card, Pencil.

Differentiation & language support

- Reminder: an outlier pulls the mean but barely moves the median (SPED).
- Sentence stem: “The better measure is ___ because ___.”
- ESOL: circle any value that looks far from the others.

Key vocabulary (English / Español):

Term	Español	What it means
outlier	valor atípico	a value far from the others
measure of center	medida de tendencia central	mean, median, or mode

Quick teacher check: *Listen for teams citing the outlier when choosing the median.*

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Name: _____

Date: _____

Class: _____

Your goal: *I can decide whether the mean or median better describes a data set and justify it.*

What you need

Data-set evidence cards, Court argument organizer.

How to play

1. Read a data set and look for an outlier.
2. Decide if the mean or median is the better measure.
3. Write your verdict and the evidence.
4. Defend your choice to the class.

Data Evidence Cards

Decide if the mean or median better describes each data set. Watch for outliers.

1. 10, 12, 14, 15, 100	2. 20, 22, 24, 26
3. Salaries: 30k, 32k, 35k, 200k	4. Test scores: 85, 88, 90, 92, 95
5. 3, 4, 5, 6, 50	6. 7, 8, 8, 9, 10
7. House prices with one mansion in the list	8. 2, 2, 3, 3, 4

Court organizer (measure + evidence):

Explain your thinking

Explain how an outlier affects the mean more than the median.

Sentence frame: *The better measure is ___ because ___.*

★ **Challenge:** For Card 1, find the mean with and without the outlier. How much does it change? (30.2 vs 12.75)

ANSWER KEY

Teacher copy — please do not distribute to students.

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Answers

1. 10, 12, 14, 15, 100 Answer: Median (14) — outlier 100 pulls the mean up	2. 20, 22, 24, 26 Answer: Mean (23) — no outlier, evenly spread
3. Salaries: 30k, 32k, 35k, 200k Answer: Median (33.5k) — 200k is an outlier	4. Test scores: 85, 88, 90, 92, 95 Answer: Mean (90) — no outlier
5. 3, 4, 5, 6, 50 Answer: Median (5) — 50 is an outlier	6. 7, 8, 8, 9, 10 Answer: Mean (8.4) — no outlier
7. House prices with one mansion in the list Answer: Median — the mansion is an outlier	8. 2, 2, 3, 3, 4 Answer: Either works (mean 2.8) — no outlier

Teaching notes & look-fors

- Use the median when there is an outlier; the mean otherwise.
- Card 1: mean is 30.2 with the 100, but 12.75 without it.